Test Specification

Assignment 4 **T**eam 2

1 Requirement 1

Requirement: When requesting a delivery, the user shall be able to specify the target room inside the office environment or staff members through a commterminal interface.

1.1 Test

Test Description: The delivery command lineinterface for the sender and receiver clients provides separate input fields for specifying their respective rooms. During testing, the inputs were entered and successfully transmitted to the delivery server.

1.2 Metrics

Metrics:

- **Input acceptance rate:** Percentage of valid room identifiers correctly accepted and logged.
- **Transmission success:** Confirmation that the delivery request is received by the server.

1.3 Justification

Justification: This requirement is strictly about functional input capability. If inputs are accepted and delivered to the server, the requirement is met. No performance or safety metric is necessary here.

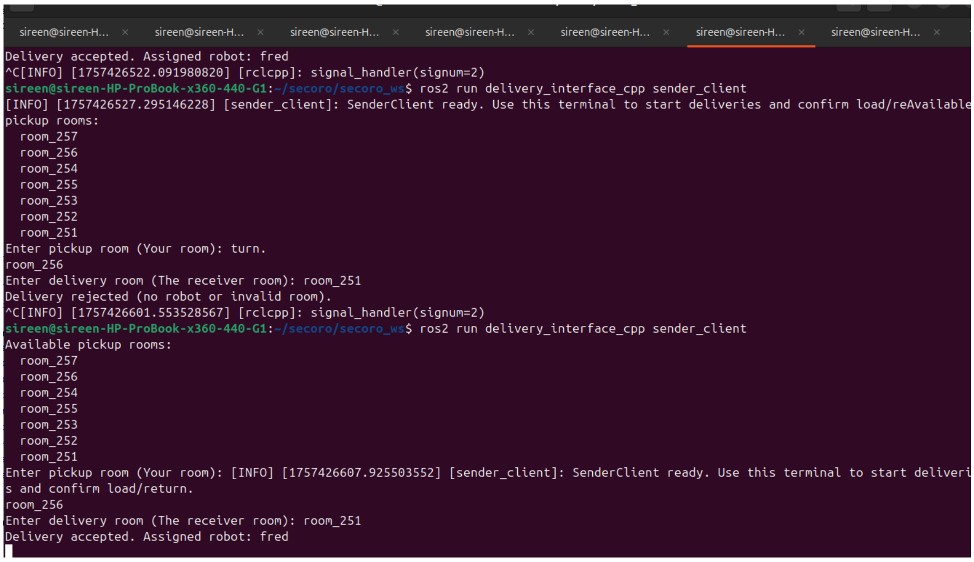

Figure 1: Interface for specifying sender and receiver rooms

2 Requirement 2

Requirement: When a delivery request is initiated, the server is supposed to manually assign one amongst the two robots, if none of them are free, it should give a message to the sender.

2.1 Test

Test Description: As soon as the sender initiates the delivery and enters the pickup room, one amongst the two robots are assigned for delivery and if the first one is busy the second one is assigned and a delivery not possible message in case of unavailabilty of both robots.

2.2 Metrics

Metrics:

- **Correct robot selection rate:** selects one amongst the 2 robots 100% times in N number of trials.
- **Notification confirmation:** Log message at the server indicating which robot has accepted the request.

2.3 Justification

Justification:
The requirement demands correct automaticc assignment to the robots.Achieving 100% correctness confirms compliance.

3 Requirement 3

Requirement: The Lidar and optical system of the robot should detect static and dynamic obstacles that are at least 10 cm high and 5 cm wide.

3.1 Test

Test Description: The system relies on the collision_monitor node, which simulates the robot's commanded velocity (cmd_vel_smoothed) forward in time for 1.2 seconds with a simulation step of 0.1s. If the predicted path intersects with an obstacle in the costmap, the robot halts. In testing, the robot successfully stopped for all qualifying obstacles.

3.2 Metrics

Metrics:

- **Detection success rate:** 100% detection of static and dynamic obstacles.
- **Reaction mechanism verification:** Confirmation that stop behavior is triggered by collision_monitor.

3.3 Justification

Justification: The requirement is explicitly about obstacle detection and avoidance. Detection success ensures obstacles of the required minimum size are never missed. Using collision_monitor guarantees detection is proactive (predictive stopping rather than reactive collision). 100% detection satisfies the requirement.

4 Requirement 4

Requirement: The robots shall autonomously navigate the indoor office environment.

4.1 Test

Test Description: Using ROS 2 navigation stack and the coded user interface, the robot autonomously navigated to specified destinations without manual intervention.

4.2 Metrics

Metrics:

- **Autonomous navigation success rate:** Percentage of trials in which the robot reached the destination without manual control (100% in tests).

4.3 Justification

Justification: The essence of this requirement is autonomy. If navigation is consistently achieved without human input, the requirement is fulfilled. Navigation efficiency (time/path length) is secondary and not explicitly required.

5 Requirement 5

Requirement: When returning the package and the sender's location is also inaccessible, the robot shall generate an incident report and notify both the sender and the assigned supervisor responsible for system monitoring.

5.1 Test

Test Description: In both cases of inaccessibility, the robot immediately generated and sent a message to the server upon detection of the blockage. The server then displayed the warning for the supervisor.

5.2 Metrics

Metrics:

- **Notification latency:** Time between detection and message generation (immediate = $<$1s).
- **Consistency rate:** 100% correct notifications over trials.

5.3 Justification

Justification: These requirements are centered on timely and correct communication. Metrics confirm that:

- The message is immediate (latency check).
- The right entity receives the message (recipient verification).
- The behavior is reliable (consistency rate).

6 Requirement 6

Requirement: When reaching the destination, the robot shall inform the sender that it has arrived and wait for the receiver to take the package.

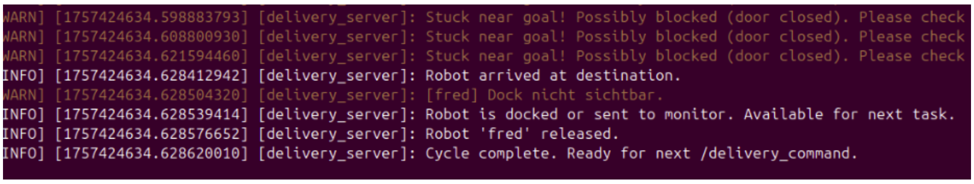

Figure 2: Incident report notification when sender is inaccessible

7 Requirement 7

Requirement: After taking the package, the receiver shall be notified by the robot to confirm receipt via the command lineinterface.

8 Requirement 8

Requirement: Upon receiving the confirmation, both the robot and the sender shall be notified, and the robot shall become available to accept a new delivery request.

8.1 Test

Test Description: These requirements were validated through the command lineinterface. At arrival, the sender received a notification. After the package was taken, the receiver was prompted to confirm receipt. Upon confirmation, both sender and robot received the final notification, and the robot's state was reset to "available."

8.2 Metrics

Metrics:

- **Notification sequence verification:** Correct order of events (arrival → receipt request → confirmation → availability).
- **Confirmation success rate:** 100% successful confirmation of receipt through the UI.

8.3 Justification

Justification: The requirement chain is about communication sequence and system state transition. Ensuring that all messages occur in the correct order and are delivered reliably proves compliance.

9 Requirement 9

Requirement: the battery level of both the robots should be visible to the users at all time, when battery level drops below a defined threshold i.e 25%, the robot shall switch to unavailable state and notify the monitoring supervisor.

9.1 Test

Test Description: Threshold was configured to 25%. When battery dropped below this level, the robot became unavailable for deliveries and notified the users. The battery percentage was displayed at all times to the users before the delivery process is initiated.

9.2 Metrics

Metrics:

- **Battery threshold trigger:** Verified at exactly 25%.
- **State transition:** Robot unavailable
- **Supervisor notification:** Log message sent immediately upon threshold trigger.

9.3 Justification

Justification: The requirement specifies both a threshold value and notification. Triggering at exactly 20% ensures adherence, while docking and notifying the supervisor confirms that safety and monitoring requirements are satisfied.

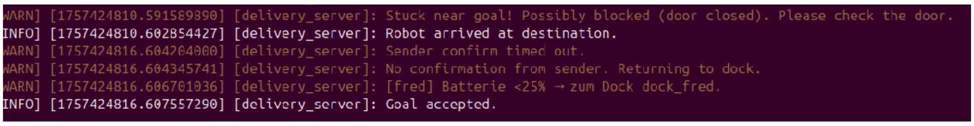

Figure 3: Battery monitoring and supervisor notification

10 Requirement 10

Requirement: list of safety briefings

10.1 Test

Test Description: a list of safety breifing is to be provided as soon as the user logs in and tries to use the services.

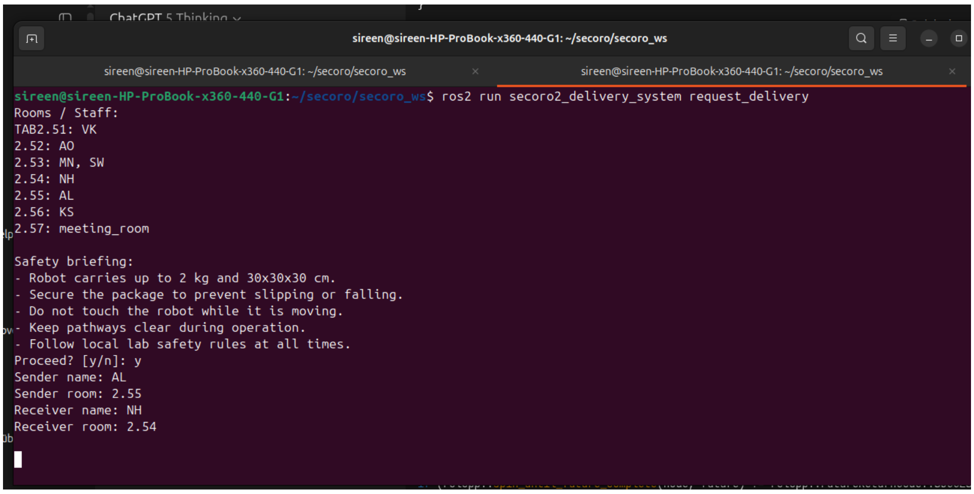

Figure 4: safety briefing

11 Requirement 11

Requirement: the complete system should be secure i.e. the users(sender and receiver should provide a username and PIN to get acces to the robot)

11.1 Test

Test Description: The users are asked for personal username and PIN before initiating the delivery through the command line interface.

11.2 Justification

Justification: Personalized usernames and PIN numbers makes it sucure for the users to use the robot.

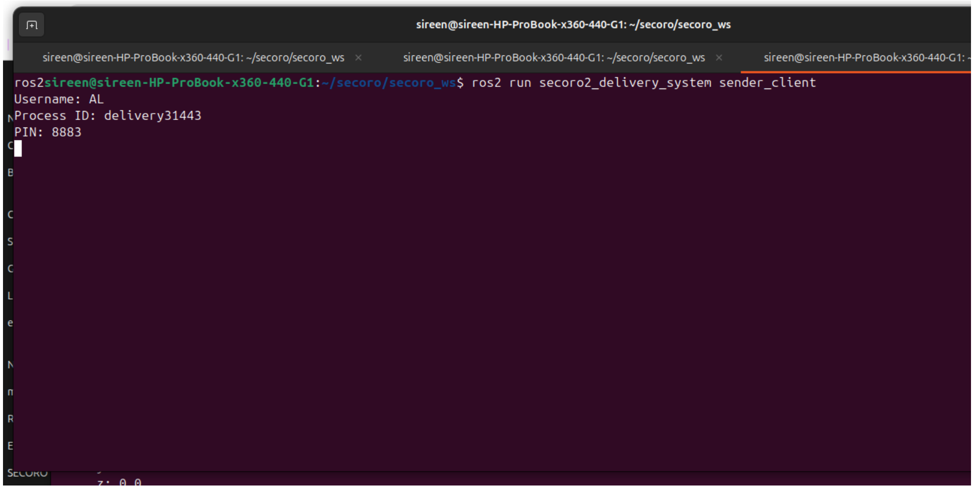

Figure 5: Security of the system